SELECT with various clauses – GROUP BY, HAVING, ORDER BY

```
mysql> desc student;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| Student_ID | int         | NO   | PRI | NULL    |       |
| Name       | varchar(30) | YES  |     | NULL    |       |
| Department | varchar(20) | YES  |     | NULL    |       |
| Age        | int         | YES  |     | NULL    |       |
| Marks      | int         | YES  |     | NULL    |       |
| City       | varchar(20) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)

mysql> SELECT * FROM Student;
+------------+--------+------------+------+-------+------------+
| Student_ID | Name   | Department | Age  | Marks | City       |
+------------+--------+------------+------+-------+------------+
|        101 | Arun   | CSE        |   20 |    85 | Chennai    |
|        102 | Bala   | ECE        |   21 |    75 | Madurai    |
|        103 | Charan | CSE        |   19 |    90 | Chennai    |
|        104 | Divya  | IT         |   20 |    65 | Coimbatore |
|        105 | Eswar  | ECE        |   22 |    80 | Salem      |
|        106 | Farah  | IT         |   21 |    95 | Madurai    |
|        107 | Gokul  | CSE        |   20 |    70 | Salem      |
|        108 | Hari   | MECH       |   23 |    60 | Chennai    |
+------------+--------+------------+------+-------+------------+
8 rows in set (0.00 sec)
```

```
mysql> SELECT Department, COUNT(*) AS Total_Students
    -> FROM Student
    -> GROUP BY Department;
+------------+----------------+
| Department | Total_Students |
+------------+----------------+
| CSE        |              3 |
| ECE        |              2 |
| IT         |              2 |
| MECH       |              1 |
+------------+----------------+
4 rows in set (0.00 sec)

mysql> SELECT Department, AVG(Marks) AS Average_Marks
    -> FROM Student
    -> GROUP BY Department
    -> HAVING AVG(Marks) > 75;
+------------+---------------+
| Department | Average_Marks |
+------------+---------------+
| CSE        |       81.6667 |
| ECE        |       77.5000 |
| IT         |       80.0000 |
+------------+---------------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Student
    -> ORDER BY Marks ASC;
+------------+--------+------------+------+-------+------------+
| Student_ID | Name   | Department | Age  | Marks | City       |
+------------+--------+------------+------+-------+------------+
|        108 | Hari   | MECH       |   23 |    60 | Chennai    |
|        104 | Divya  | IT         |   20 |    65 | Coimbatore |
|        107 | Gokul  | CSE        |   20 |    70 | Salem      |
|        102 | Bala   | ECE        |   21 |    75 | Madurai    |
|        105 | Eswar  | ECE        |   22 |    80 | Salem      |
|        101 | Arun   | CSE        |   20 |    85 | Chennai    |
|        103 | Charan | CSE        |   19 |    90 | Chennai    |
|        106 | Farah  | IT         |   21 |    95 | Madurai    |
+------------+--------+------------+------+-------+------------+
8 rows in set (0.00 sec)

mysql> SELECT * FROM Student
    -> ORDER BY Marks DESC;
+------------+--------+------------+------+-------+------------+
| Student_ID | Name   | Department | Age  | Marks | City       |
+------------+--------+------------+------+-------+------------+
|        106 | Farah  | IT         |   21 |    95 | Madurai    |
|        103 | Charan | CSE        |   19 |    90 | Chennai    |
|        101 | Arun   | CSE        |   20 |    85 | Chennai    |
|        105 | Eswar  | ECE        |   22 |    80 | Salem      |
|        102 | Bala   | ECE        |   21 |    75 | Madurai    |
|        107 | Gokul  | CSE        |   20 |    70 | Salem      |
|        104 | Divya  | IT         |   20 |    65 | Coimbatore |
|        108 | Hari   | MECH       |   23 |    60 | Chennai    |
+------------+--------+------------+------+-------+------------+
8 rows in set (0.00 sec)
```